Pinpoint.

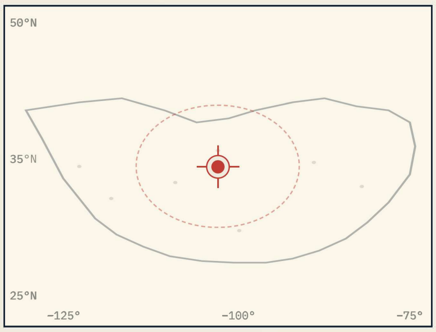
Can a coarse GPS prior narrow the satellite search space enough that cross-view retrieval delivers street-level geolocalization *— without ballooning the compute budget?*

GROUP 14
M. Andersen · K. Larsen
S. Nakamura · J. Pereira

Multi-Modal Geo-Spatial
Learning · MMLandmarks

— TWO-STAGE DESCENT · GROUND PHOTO → GPS → SATELLITE TILE —

/01 CONTINENTAL SCALE · 10° GRID



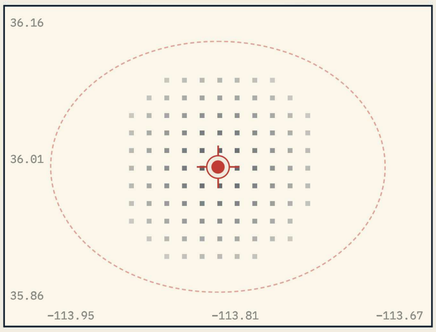
GeoCLIP × multi-image

Frozen CLIP image tower + learned attention pool over the landmark's photos predicts a single (lat, lon).

prior lat 36.012 · lon -113.811
σ ±20 km radius

→ CROP ~20 KM

/02 NARROW · ~100 CANDIDATE TILES



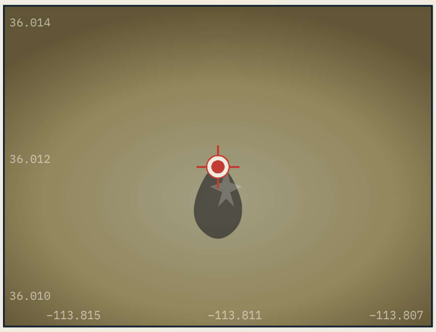
GPS-guided *narrowing*

Haversine filter against the 101 K-tile satellite index keeps only the ~100 closest candidates per query.

101 K → 102 tiles
reduction 1000 ×

→ RE-RANK

/03 TILE · 100 M RESOLUTION



Sample4Geo *match*

Siamese ConvNeXt-B (shared weights, symmetric InfoNCE) scores ground ↔ each candidate; top-1's GPS is the final prediction.

final 36.0120, -113.8111
err < 100 m

§ I THE DATA

MMLandmarks: *four* views of a US landmark.

An instance-level cross-view benchmark — every landmark carries a ground photo, an aerial tile, a Wikipedia paragraph, and a GPS coordinate, jointly indexed.

GROUND PHOTO | SATELLITE | GPS | WIKI TEXT

17 K TRAIN LANDMARKS

311 K TRAIN GROUND IMGS

101 K SATELLITE TILES

18 K QUERY IMAGES · 1 K LANDMARKS

§ II WHY BRIDGE GPS IN?

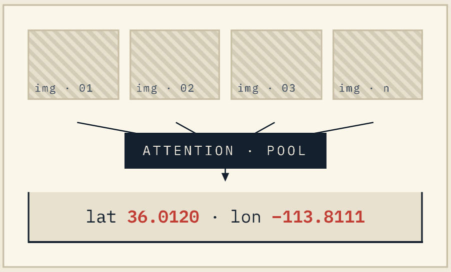
Cross-view retrieval against 101 K satellite tiles is *O(N) and noisy*. A cheap GPS prior collapses the gallery to ~10^2 candidates per query — visually and geographically similar ones — turning a global match into a tractable local re-ranking.

GeoCLIP is wide & coarse. Sample4Geo is narrow & fine. Combined, they recover the spatial resolution of the satellite tile (meters) instead of the location encoder's (kilometers).

§ III THE NOVELTY

Many photos, *one* landmark — attention-pooled GeoCLIP.

Stock GeoCLIP scores one image at a time. We re-cast it as a set encoder: an encoder-only transformer pools the CLIP-image features of *all* photos of a landmark into a single embedding before contrastive alignment with GPS.



+6.65 pp Acc@1 km and **−185 km** median error vs. zero-shot single-image.

§ IV JOINT LOSS

$$\mathcal{L}_{total} = \alpha \cdot \mathcal{L}_{gps} + \beta \cdot \mathcal{L}_{sat}$$

Symmetric InfoNCE on both branches — image↔location for the prior, ground↔satellite for the matcher — coupled through the shared backbone.

§ V RESULTS

CONFIGURATION	R@1	R@5	R@10
— GEOCLIP · GROUND → GPS · ACC@1 KM			
Zero-shot off-the-shelf	6.67%	28.79%	44.48%
Multi-image · mean softmax-cos	12.13%	40.67%	58.45%
Multi-image · attention ours	13.32%	54.46%	72.63%
— SAMPLE4GEO · GROUND → SATELLITE · R@K			
ConvNeXt zero-shot ImageNet only	0.25%	0.76%	1.13%
v2 · 224 px InfoNCE + DSS	7.21%	18.52%	25.31%
v3 · 384 px multi-positive	8.58%	18.13%	22.29%
— HYBRID PIPELINE · GEOCLIP → SAMPLE4GEO			
ZS GeoCLIP + v3 fixed prior	10.20%	21.11%	26.80%
Finetuned hybrid α=β=1	13.80%	22.23%	25.41%
★ **Multi-img + hybrid** full system	**15.50%**	–	–

§ VI WHAT WE LEARNED

1. **Pretraining matters more than tuning.** ConvNeXt's ImageNet-only zero-shot scored 0.25 % R@1; one fine-tune pass cleared 8 %.
2. **Resolution buys context.** 384 px beats 224 px by +1.37 pp R@1.
3. **Set-encoding works.** Attention pooling doubles Acc@1 km over zero-shot single-image GeoCLIP.
4. **GPS narrowing pays off.** Hybrid + multi-image lifts R@1 by **+6.9 pp** over standalone v3.

§ VII WHAT WE'D TRY NEXT *four extensions, ranked by impact ÷ effort.*

→ 01 **Adaptive radius.** Use GeoCLIP's softmax temperature as a confidence signal; widen R when uncertain.

→ 02 **Wiki as a third anchor.** Late-fuse the CLIP text encoding of the landmark's article into the re-ranker.

→ 03 **Per-landmark eval.** Reweight metrics so popular tourist sites stop dominating the headline number.

→ 04 **Uni-1652 transfer.** Test whether the pipeline generalizes off-continent.

References · MMLandmarks [Kristoffersen et al., arXiv:2512.17492 · 2025] · Sample4Geo [Deuser et al., ICCV 2023] · GeoCLIP [Vivanco Cepeda et al., NeurIPS 2023] · CLIP [Radford et al., 2021] · Earth Embeddings [Klemmer et al., 2025]

15.50%
Acc@1km · full system